LCD + Serpentine Optical System

3D Free-Space Diffraction & Fringe Imaging — Experiment Summary

1. Overview

This document summarizes a series of optical experiments in which a cracked LCD panel was rotated within the free space between a Serpentine waveguide output aperture and a convex lens. Two aperture geometries were tested: standard Serpentine and Honeycomb Serpentine. The primary goal was to observe diffraction, fringe, and interference behavior — particularly any 3D spatial image formation analogous to holographic projection.

2. Experimental Setup

Component	Description
Light Source	Serpentine waveguide output aperture
Optical Element	Cracked LCD panel (inserted and rotated)
Focusing Element	Convex lens
Aperture Geometry A	Standard Serpentine
Aperture Geometry B	Honeycomb Serpentine
Observation Space	Free space between aperture and lens + screen plane
Recording	Video (MP4), 5 clips per geometry

3. Experiment A — Standard Serpentine

Date: 2026-05-28 | Files: 20260528_205003 ~ 205331 (5 clips)

Key Observations:

• No satellite peaks detected — only fringe lines traversing the field.

• At 0° rotation: simultaneous blinking observed along both x-axis and y-axis.

• Fringe lines pass through the field without satellite formation.

• Most notably: interference images form and move in **3D free space**, not confined to the screen plane — hologram-like behavior.

• Spatial image formation observed in open 3D space as the LCD is rotated.

Interpretation:

The Serpentine aperture appears to produce a coherent wavefront capable of forming real, spatially extended interference structures beyond the screen plane. The simultaneous x/y axis blinking at 0° suggests the aperture geometry imposes symmetrical boundary conditions on the diffracted wavefront. The absence of satellite peaks implies a clean, single-mode-like output from the aperture.

4. Experiment B — Honeycomb Serpentine

Date: 2026-05-29 | Files: 20260529_045815 ~ 050114 (5 clips)

Key Observations:

- Overall behavior consistent with Experiment A: no satellites, fringe-only diffraction, x/y blinking at 0°.
- Central ring is **more prominent** compared to standard Serpentine.
- Fringe structures are **more discrete** and spatially well-defined in 3D free space.
- 3D holographic-like image formation effect persists and appears enhanced.
- Spatial coherence of interference pattern visibly improved with Honeycomb geometry.

Interpretation:

The hexagonal symmetry of the Honeycomb Serpentine geometry appears to enhance wavefront coherence, resulting in a sharper and more spatially resolved fringe pattern. The more prominent central ring and increased discreteness of fringes suggest improved mode confinement or aperture-level phase coherence compared to standard Serpentine. This supports the hypothesis that aperture geometry directly governs the spatial structure of the emitted wavefront.

5. Comparative Summary

Parameter	Standard Serpentine	Honeycomb Serpentine
Satellite peaks	Absent	Absent
Fringe lines	Present	Present (more discrete)
X/Y blinking at 0°	Yes	Yes
Central ring	Visible	More prominent
3D free-space imaging	Yes (hologram-like)	Yes (enhanced)
Fringe discreteness	Moderate	High
Spatial coherence	Good	Improved

6. Conclusions

1. Both Serpentine geometries produce coherent, hologram-like 3D fringe imaging in free space — without satellites.
2. The 0° angle consistently triggers simultaneous x and y axis blinking, indicating aperture-imposed symmetry.
3. Honeycomb Serpentine geometry yields superior spatial coherence: sharper central ring and more discrete fringes.
4. The aperture geometry is a critical parameter governing wavefront structure and 3D interference pattern quality.
5. Results suggest the system may be capable of true wavefront shaping for holographic or beam-structuring applications.

7. Next Steps

- Quantitative fringe spacing analysis across rotation angles.
- Test additional aperture geometries (e.g. triangular, spiral Serpentine).
- Measure 3D spatial extent of interference image as a function of lens focal length.
- Compare behavior with intact (non-cracked) LCD panels.
- Explore potential applications in holographic display or structured light generation.